UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

DELIVERABLE No. 2: METHOD FIT MATRIX

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Scoped research question: How should a reference architecture based on specialized generative AI agents be structured to automate and verify the development of critical state information systems, ensuring their compliance with the enterprise architecture and applicable regulations?

Criterion / Weight	Pure Experiment	Simulation	Mixed Methods	Design Science Research
E: Epistemological Fit (30%)	**3/5 -** Can test causal effects (e.g., whether separating generation and verification lowers defects), but does not govern the design of the reference-architecture artifact.	**2/5 -** Can model the agent pipeline and inject faults, yet fits weakly with a knowledge claim about a real, governed artifact under enterprise-architecture and regulatory constraints.	**4/5 -** Integrates quantitative and qualitative evidence, but is broader than-and does not itself produce-the core design contribution.	**5/5 -** Natural fit: knowledge is generated by building and evaluating the reference-architecture artifact.
D: Data Feasibility (20%)	**2/5 -** Randomized, comparable development tasks in a live public entity are hard to obtain, and sensitive pension data constrain access.	**3/5 -** Synthetic workloads and injected defects are possible, but faithful models of the enterprise architecture and regulations are resource-intensive.	**3/5 -** Requires both repeated quantitative measurements and qualitative expert data.	**4/5 -** Prototype logs, enterprise-architecture documentation, code repositories, development artifacts, and focused expert feedback are obtainable without personal data.
F: Feasibility (15%)	**2/5 -** High control and randomization burden, plus production-like access, are hard to secure within three years.	**3/5 -** Feasible, but faithful modeling of regulations, EA rules, and correlated failure modes adds time.	**3/5 -** Feasible only if tightly scoped; a full mixed-methods study risks overextension.	**4/5 -** Feasible if limited to a prototype, a specific context (the pension system or possibly the Peruvian educational scholarship system), a scenario-based evaluation, and 3–4 key metrics.
C: Contribution (20%)	**2/5 -** Produces effect evidence on isolated variables, not the reference-architecture artifact or transferable design knowledge.	**2/5 -** Explains behavior under modeled conditions but does not deliver the governed artifact or real-world design principles.	**4/5 -** Produces integrated evidence, but artifact/design knowledge is not central.	**5/5 -** Delivers the artifact (reference architecture) + rigorous evaluation + transferable design principles (conformance principle; generation/verification separation).
V: Venue Fit (15%)	**3/5 -** Publishable in empirical-SE / evaluation venues if the causal design is defensible, but the design contribution is the intended story.	**3/5 -** Possible in modeling / dependability venues; weaker match to the design and governance contribution.	**4/5 -** Well accepted in empirical-SE and information-systems venues, but with broader study burden.	**5/5 -** Strong fit with ECSA, IEEE Software, Journal of Systems and Software, and Government Information Quarterly (public-sector angle).
Weighted Total (E×.30 + D×.20 + F×.15 + C×.20 + V×.15)	**2.45/5 (49/100)**	**2.50/5 (50/100)**	**3.65/5 (73/100)**	**4.65/5 (93/100)**

Decision: Design Science Research is selected as the primary method. Pure Experiment and Simulation are eliminated by the ≤2 gate on a critical criterion (Contribution for both; also Epistemological Fit for Simulation), because neither can produce or govern the reference-architecture artifact that is the intended contribution. Mixed Methods remains a viable evaluation strategy embedded within DSR-not the governing method-consistent with the paradigm justification. To stay feasible within three years, the DSR study is restricted to one functional prototype (the generation/verification agent pipeline with a conformance layer), one application context (the Peruvian pension system or possibly the Peruvian educational scholarship system), a scenario-based architecture evaluation with expert participation, and 3–4 primary outcome metrics. A pure experiment may be embedded inside the DSR evaluation cycle as a controlled comparison of the two-agent pipeline against a single-agent baseline, and simulation may be used for pre-deployment stress testing of defect-injection and correlated generator–verifier failure scenarios. Color legend: red = weak fit (≤2); amber = moderate fit (3); green = strong fit

(≥4); blue = structural rows.

Gate rule: any candidate scoring ≤2 on a critical criterion (Epistemological Fit or Contribution) is removed. Simulation is not shortlisted as the primary method because the central claim concerns real-world utility and governed conformance in a regulated public system; it may still be used for pre-deployment stress testing of failure scenarios (e.g., injecting defects and correlated generator–verifier errors).

Data sources for the selected DSR plan: prototype interaction and build logs; automatically detected conformance violations (precision/recall) against enterprise-architecture and regulatory rules; defect escape rate, rework effort, and lead time; the single-agent (no independent verification) baseline; enterprise-architecture documentation, code repositories, and development artifacts from the application case; and focused feedback from architecture and domain experts. Prerequisites: alignment with Law No. 29733 (Protection of Personal Data), Law No. 31814, and NTP-ISO/IEC 42001:2025; restriction of the artifact to development and verification (not production execution); data minimization with no identifiable personal pension data; and local / on-device processing with access limited to strictly necessary code and data.